Code No: 134BD **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech II Year II Semester Examinations, September/October - 2023
FORMAL LANGUAGES AND AUTOMATA THEORY
(Common to CSE, IT)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

PART – A

(25 Marks)

1.a)	What is δ in NFA with ϵ moves?	[2]
b)	Define DFA.	[3]
c)	State pumping lemma for regular languages.	[2]
d)	Give recursive definition for Regular Expression.	[3]
e)	What is parse tree?	[2]
f)	What do you mean by PDA with empty Stack?	[3]
g)	Write Instantaneous Descriptions for the Turing Machines.	[2]
h)	Write about the programming techniques for Turing Machines.	[3]
i)	What are recursively enumerable languages?	[2]
j)	Define undecidable problem.	[3]

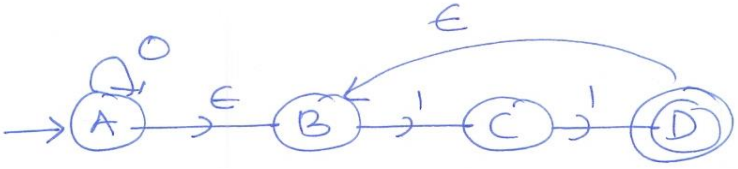
PART – B

(50 Marks)

2.a) Design DFA to accept strings with c and d such that number of d's are divisible by 3
b) Construct a DFA accepting the set of all strings ending with 00? [5+5]

OR

3. Convert the following NFA with € moves to DFA. [10]



4.a) State Pumping Lemma for Regular Sets. Apply pumping lemma for the language $L= \{a^n b^n / n >=1\}$ and prove that L is not regular.
b) What are the applications of regular expressions? [5+5]

OR

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5. State Myhill Nerode Theorem. Minimize the following DFA using Myhill Nerode Theorem. [10]

S	0	1
→ A	B	C
B	D	C
C	B	E
(D)	D	E
(E)	D	E

6.a) Show that the following grammar is ambiguous.

E →E + T / T

T→ T*F / F

F→(E) / id

b) Consider the following grammar G = ({S,A}{a,b}, P, S) Where P consists of S→ aAS |a, A→ SbA | SS | ba. For the string **aabbaa** show

i) Left most derivation ii) Right most derivation. [5+5]

OR

7. Design PDA for the Language L= {WcW^R / W is in (0+1)*} [10]

8. What is Griebach Normal Form (GNF)? Reduce the following CFG into GNF.
S→SS, S→0S1/01 [10]

OR

9. Design a TM for the language $L = \{1^n 0^n \mid n \geq 1\}$ [10]

10. Explain the following
a) Class of P Problem
b) NP- Complete Problem. [5+5]

OR

11. Explain about
a) Universal Turing Machine
b) Modified Post Correspondence Problem. [5+5]

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